MATHEMATICS

Analysis and approaches

Standard Level | Set B | Paper 2

Mock examination · 2025 series

Time allowed: 1 hour 30 minutes Maximum mark: 80

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A graphic display calculator is required.

• This paper has Section A and Section B. Both sections are compulsory.

• Answer every question. Show all working and give sufficient mathematical reasoning.

• A clean Mathematics: analysis and approaches formula booklet may be used.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures unless instructed differently.

• Diagrams are not necessarily drawn to scale. Use the stated measurements and conditions.

• The number in square brackets shows the maximum mark for each part.

SECTION A

Question 1 [Maximum mark: 4]

Events A and B are independent, with $P(A) = 0.34$ and $P(B) = 0.45$.

(a) Find $P(A \cap B)$. [2]

(b) Find $P(A \cup B)$. [2]

SECTION A

Question 2 [Maximum mark: 6]

Five paired measurements are shown. The x-measurements are integers. The regression line of y on x, with its coefficients rounded to 3 decimal places, is $y = 2.398x + 101.446$.

x	12	15	17	a	28
y	127	142	144	151	171

(a) Find $\bar{y}$. [1]

(b)(i)) Use the regression line to estimate $\bar{x}$, giving your answer to the nearest whole number. [2]

(b)(ii)) Hence find a. [2]

(c) Find the Pearson correlation coefficient r. [1]

SECTION A

Question 3 [Maximum mark: 5]

Let $f(x) = 2x^2 + 20x + 54$, restricted to $x \leq -5$.

(a) The range is $y \geq h$. Find h. [1]

(b) Find $f^{-1}(x)$. [4]

SECTION A

Question 4 [Maximum mark: 6]

A fair coin is tossed 12 times independently. Let X be the number of tails.

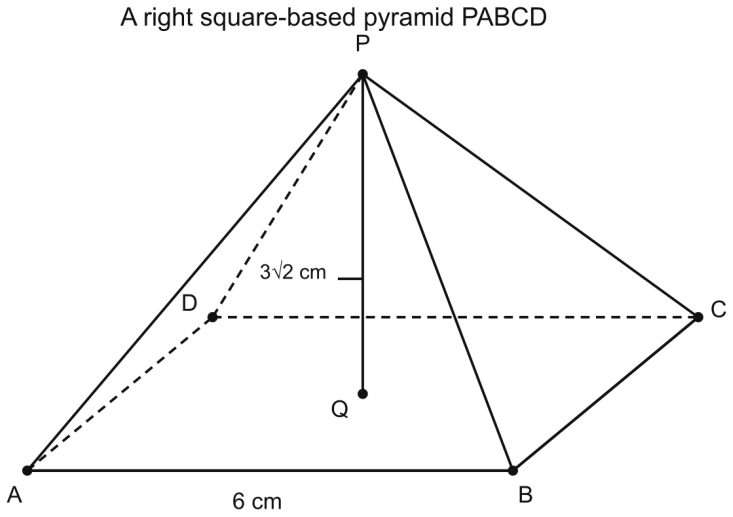
(a) State the distribution of X and find the probability of obtaining equal numbers of heads and tails. [3]

(b) Find the probability of obtaining more tails than heads. [3]

SECTION A

Question 5 [Maximum mark: 7]

A right square-based pyramid $PABCD$ has base side 6 cm. The apex P is vertically above the centre Q of the square, and $PQ = 3\sqrt{2}$cm.



Q is the centre of the square base; PQ is perpendicular to its plane.

(a) Find the exact volume of the pyramid. [2]

(b) Find the exact length AC. [2]

(c) Show that triangle PAB is equilateral. [3]

SECTION A

Question 6 [Maximum mark: 6]

Consider $\cos(\pi x/10) = 0$ on the interval $0 \leq x \leq 1000$.

(a) Find all solutions, using a formula with an integer parameter. [3]

(b) Find the sum of all these solutions. [3]

SECTION A

Question 7 [Maximum mark: 6]

Let $f(x) = \frac{3}{2}x^4 - \frac{1}{3}x^3 - \frac{5}{2}x^2 + 83$.

(a) Find $f'(x)$. [2]

(b) Find $f'(-1)$. [1]

(c) Find the other two x-coordinates at which the tangent is parallel to the tangent at $x = -1$. [3]

SECTION B

Question 8 [Maximum mark: 14]

Two fair six-sided dice are rolled independently. Let D be the absolute difference of their scores.

(a) Find $P(D = 0)$. [1]

(b) Find $P(D = 5)$. [2]

(c) Find $P(D = 4)$. [3]

(d) Complete the following probability table by finding x, y, z.

d	0	1	2	3	4	5	
$P(D = d)$	$6/36$	$10/36$	$x/36$	$6/36$	$y/36$	$z/36$	[2]

(e) Find $E(D)$. [2]

(f) A game gives a net profit of 7 dollars if $D = 5$, a net profit of 1 dollar if $D = 1$, and a net loss of L dollars otherwise. Find L if the game is fair. [4]

SECTION B

Question 9 [Maximum mark: 13]

The mass R of a standard packet is normally distributed with mean 500 g and standard deviation 18 g. Packets are sampled independently. A premium variety has a normally distributed mass C, with lower quartile 724 g and interquartile range 52 g.

(a) Find the probability that a standard packet has mass less than 480 g. [1]

(b) Given that a standard packet has mass greater than 480 g, find the probability that it has mass less than 500 g. [2]

(c) Find a such that 15% of standard packets have mass greater than a grams. [2]

(d) Five standard packets are selected. Find the probability that exactly one has mass less than 480 g. [3]

(e)(i)) Find the mean of the premium mass distribution. [2]

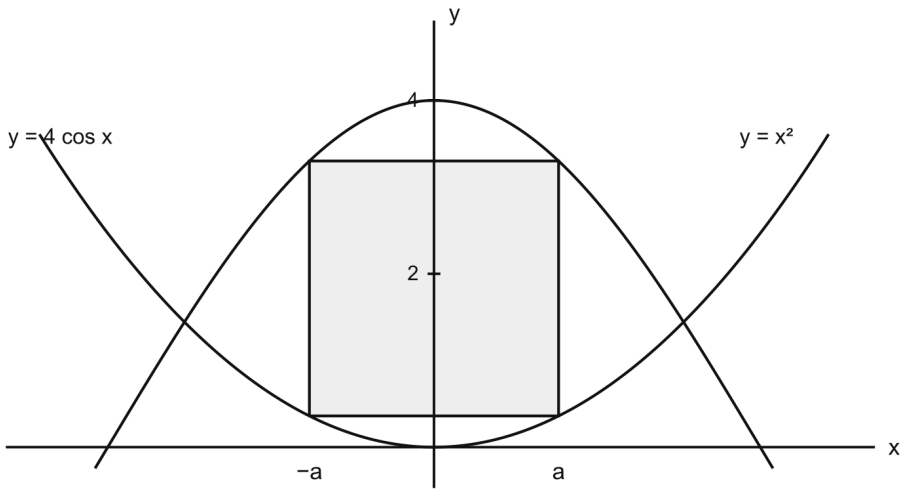
(e)(ii)) Find the standard deviation of the premium mass distribution. [3]

SECTION B

Question 10 [Maximum mark: 13]

The curves $y = 4\cos x$ and $y = x^2$ intersect at L and R, with R having positive x-coordinate r. Angles are in radians. A rectangle has vertical sides at $x = -a$ and $x = a$, its upper vertices on $y = 4\cos x$ and lower vertices on $y = x^2$, where $0 < a < r$. Its area is A.

A rectangle between two curves; angles in radians



(a) Find the coordinates of L and R. [2]

(b) Find the total area enclosed by the two curves. [4]

(c) Show that $A = 8a\cos a - 2a^3$. [2]

(d)(i)) Show that a stationary value of A satisfies $\frac{\cos a - a\sin a}{a^2} = \frac{3}{4}$. [3]

(d)(ii)) Find the maximum possible area of the rectangle. [2]

End of question paper

MATHEMATICS

Analysis and approaches

Standard Level | Set B | Paper 2

Worked solution book

This section contains the worked solutions and marking points. Use it after completing the question paper. Award method, accuracy and reasoning marks separately, and accept mathematically equivalent valid methods.

Mark your written working against the solution book before entering self-assessed marks in the portal. A correct final answer does not always earn every method or proof mark.

Section A

1. [Maximum mark: 4]

(a) $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B) = 0.34 \times 0.45 = 0.153$. **M1 A1**

[2 marks]

(b) $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B) = 0.34 + 0.45 - 0.153 = 0.637$. **M1 A1**

[2 marks]

(a) $\overline{y} = 147$. **A1**

[1 mark]

(b) (i) the point $(\overline{x}, \overline{y})$ is on the regression line, hence

$\overline{y} = 2.398\overline{x} + 101.446 \Leftrightarrow$ **M1**

$\Leftrightarrow 147 = 2.398\overline{x} + 101.446$

$\Leftrightarrow \overline{x} = 19$ **A1**

(ii) **EITHER** by trial and error

OR by $\overline{x} = 19 = \frac{12+15+17+a+28}{5} \Leftrightarrow a = 23$. **M1 A1**

[4 marks]

(c) Find the correlation coefficient $r = 0.964$. **A1**

[1 mark]

$f(x) = 2x^2 + 20x + 54$, $x \leq -5$.

(a) Vertex at $(-5, 4)$ (by GDC graph)

Vertex form: $f(x) = 2(x+5)^2 + 4$

$h = 4$ **A1**

[1 mark]

(b) $f(x) = 2(x+5)^2 + 4 = y$ **M1**

$(x+5)^2 = \frac{y-4}{2}$ **A1**

$x + 5 = -\sqrt{\frac{y-4}{2}}$ **A1**

$x = -5 - \sqrt{\frac{y-4}{2}}$

$f^{-1}(x) = -5 - \sqrt{\frac{x-4}{2}}$ **A1**

[4 marks]

(a) Binomial distribution with $n = 12$, $p = \frac{1}{2}$ **A1**

$\mathrm{P}(X = 6) = 0.22558... \cong 0.226$ **M1 A1**

[3 marks]

(b) **METHOD A**

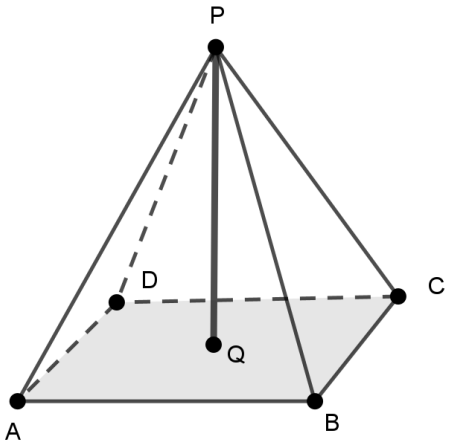
$\mathrm{P}(X \geq 7) \cong 0.387$ **M2 A1**

METHOD B

Probability of different scores: $1 - 0.22558... \cong 0.77442...$ **M1**

Probability of obtaining **more** tails than heads (due to symmetry) **M1**

$\frac{0.77442...}{2} \cong 0.387$ **A1**



(a) $Volume = \frac{1}{3} \times 6 \times 6 \times 3\sqrt{2} = 36\sqrt{2} \quad (\cong 50.9)$ **M1 A1**

[2 marks]

(b) $AC^2 = 6^2 + 6^2 = 72$ **M1**

$AC = \sqrt{72} = 6\sqrt{2} \quad (= 8.48528... \cong 8.49)$ **A1**

[2 marks]

(c) $AQ = 3\sqrt{2}$ **A1**

$PA^2 = \left(3\sqrt{2}\right)^2 + \left(3\sqrt{2}\right)^2 \Rightarrow PA = \sqrt{18+18} = 6$ **A1**

$PA = PB = 6$

Hence the triangle PAB is equilateral. **R1**

[3 marks]

$$\cos\frac{\pi x}{10}=0$$

Solving **EITHER** analytically **OR** by GDC, the solutions are

5, 15, 25, …, 995 (**OR** $x=5+10k$, with k from 0 to 99) **A2 A1**

Then

METHOD A (Arithmetic sequence) **M1**

Arithmetic sequence, $S_{100}=50,000$ **A2**

METHOD B (using Sigma notation) **M1**

$$\sum_{x=0}^{99}(5+10x)=50,000$$ **A2**

METHOD C **M1**

Using Recursion: $a_n=5+10n$, sum $=50,000$ **A2**

[6 marks]

$f(x) = \frac{3}{2}x^4 - \frac{1}{3}x^3 - \frac{5}{2}x^2 + 83$.

(a) $f'(x) = 6x^3 - x^2 - 5x$. **A1 A1**

[2 marks]

(b) $f'(-1) = -6 - 1 + 5 = -2$ **A1**

[1 mark]

(c) $f'(x) = 6x^3 - x^2 - 5x = -2 \Leftrightarrow 6x^3 - x^2 - 5x + 2 = 0$ **M1**

Remaining solutions: $x = \frac{2}{3}$, $x = \frac{1}{2}$ **A1 A1**

[3 marks]

8. [Maximum mark: 14]

(a) 6 cases: 1-1, 2-2, 3-3, 4-4, 5-5, 6-6 out of 36

$$\mathrm{P}(D=0)=\frac{6}{36}=\frac{1}{6}$$ **A1**

[1 mark]

(b) cases: 1-6, 6-1 **M1**

$$\mathrm{P}(D=5)=\frac{2}{36}\quad\left(=\frac{1}{18}\right).$$ **A1**

[2 marks]

(c) cases: 1-5, 5-1, 2-6, 6-2 **M1**

$$\mathrm{P}(D=4)=\frac{4}{36}\quad\left(=\frac{1}{9}\right).$$ **A2**

[3 marks]

(d) $x=8$ **A1**

$y=4$, $z=2$. **A1**

d	0	1	2	3	4	5
$\mathrm{P}(D=d)$	$\frac{6}{36}$	$\frac{10}{36}$	$\frac{8}{36}$	$\frac{6}{36}$	$\frac{4}{36}$	$\frac{2}{36}$

[2 marks]

(e) $$\mathrm{E}(D)=\frac{10+16+18+16+10}{36}=\frac{70}{36}=\frac{35}{18}$$ **M1 A1**

[2 marks]

(f) $$\mathrm{E}(\text{Profit})=7\times\frac{2}{36}+1\times\frac{10}{36}-L\times\frac{24}{36}$$ **M1 A1**

Fair game: $\mathrm{E}(\text{Profit})=0$ **M1**

$L=1$. **A1**

[4 marks]

(a) Let R be the variable for the weight if the packet.

$\mathrm{P}(R < 480) = 0.13326... \cong 0.133$. **A1**

[1 mark]

(b) $\mathrm{P}(R < 500 \mid R > 480) = \dfrac{\mathrm{P}(480 < R < 500)}{\mathrm{P}(R > 480)} = \dfrac{0.36673...}{0.86673...} \cong 0.423$ **M1 A1**

[2 marks]

(c) $\mathrm{P}(R > a) = 0.15 \Rightarrow a = 518.65... \cong 519$ **M1 A1**

[2 marks]

(d) Let B be the variable for the number of packets under 480

Binomial with $n = 5$ and $p = 0.13326...$ **M1 A1**

$\mathrm{P}(B = 1) = 0.376$ **A1**

[3 marks]

(e) Let C be the variable for the weight of the packet of the premium variety.

(i) $Q_3 = 724 + 52 = 776$, $\mu = \dfrac{724 + 776}{2} = 750$ **M1 A1**

(ii) $\mathrm{P}(C < 724) = 0.25$. **M1**

$Z = \dfrac{C - \mu}{\sigma} \Rightarrow Z = \dfrac{724 - 750}{\sigma} \Rightarrow Z = \dfrac{-26}{\sigma}$ **M1**

From InvN(0,1): $Z = -0.67448$

$-0.67448 = \dfrac{-26}{\sigma} \Rightarrow \sigma = 38.5$ **A1**

[5 marks]

(a) $(-1.20153...,1.44369...)$ so $(-1.20,1.44)$ **A1**

$(1.20153...,1.44369...)$ so $(1.20,1.44)$ **A1**

[2 marks]

(b) $\int_{-1.20153}^{1.20153...} \left(4\cos x - x^2\right)\mathrm{d}x \cong 6.30$ **M1 A1 A1 A1**

[4 marks]

(c) Base = $2a$, height = $h = 4\cos a - a^2$. **A1**

$A = 2a(4\cos a - a^2) = 8a\cos a - 2a^3$. **A1**

[2 marks]

(d) (i) $\dfrac{\mathrm{d}A}{\mathrm{d}a} = 8\cos a - 8a\sin a - 6a^2 = 0 \Leftrightarrow 8\cos a - 8a\sin a = 6a^2$ **M1 M1 A1**

$\Leftrightarrow \dfrac{\cos a - a\sin a}{a^2} = \dfrac{3}{4}$

(ii) $A_{\max} \cong 3.60$. [directly from the graph of A] **M1 A1**

[5 marks]